Pavel Vasilyev

LLM systems engineer · applied ML and NLP since 2009

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github.com/marchdown · dissemblage.art

I build the part of an AI product that has to be correct before it ships: provider routing and fallback, budgets and rate limits, evaluation benches, and the review process around them. Fifteen years in NLP and applied machine learning before that, and the habit of writing the specification first and leaving a repository someone else can run.

RECENT WORK, WITH WHAT IT SHOWED

LLM inference harness
ELIXIR · RUST · PRIVATE; PYTHON PART PUBLIC

Provider routing with role-specific fallback chains, jittered retry honouring `Retry-After`, per-client rate limits, bounded concurrency, and token, cost and latency audit logs per call. Measured over 88 requests: 100 % success on a quota probe, 83 % in an outage probe, p95 latency 6.4 s. *Cost, latency and quota visible per call.*

gemini-gate
ELIXIR / PHOENIX · PRIVATE

A credential service in front of the Gemini API: authenticated retrieval, constant-time token comparison, atomic credential replacement, deployed under Caddy and systemd. *Enforced boundaries around agents, not prompt rules.*

tidy-harness
SWIFT · PUBLIC

A verifier for agent behaviour: rules as data, a pure deterministic lens over session logs, and a Babashka reference that must agree with the Swift core exactly before a number is believed. They agree on 151 real sessions. *Two implementations, one answer, or no answer.*

pocket
PWA PIPELINE · LIVE, CODE PRIVATE

Eighteen installable, offline-first web apps behind one pipeline: a machine-checked rule set (offline cache, update banner, safe-area, no network on the critical path), content-hash service-worker versions, and a deploy that verifies the live URLs from outside. *End-to-end delivery, build to verified deploy.*

co-loop
PROTOCOL · PUBLIC

A fork-based protocol for parallel agentic work: several agents on one codebase without stepping on each other's files, and clean integration afterwards. *How I run more than one coding agent at once.*

dissemblage.art
TEN INTERACTIVE PIECES · LIVE

First-person rooms in hyperbolic 3-space, an aperiodic binary tiling, generative card decks that compose a system prompt or an AGENTS.md from twenty choices. Each is one HTML file with no build step. *Craft, and geometry, in public.*

topo-sonify
RUST · RESEARCH PROTOTYPE, APR–JUL 2026

Persistent homology, Hodge theory and Morse theory mapped to sound. The mathematical core was checked against known invariants ($\beta(S^2) = (1,0,1)$, $\chi = 2$) before any audio code; stopped after the WASM build. *Applied-mathematics depth behind the engineering.*

EXPERIENCE

2010 – present — Independent software engineer · consulting and open source
Design, build and ship software for clients and my own projects: 150+ projects across eight or more languages, with LLM coding agents as my main tool since 2022.

2026 — LLM research contract · Move38
Explored and tested LLM strategies for an API-centred research angle: ontologies, prompt sets, test benches and harnesses across OpenRouter, Modal and Railway.

2024 — Technical infrastructure consultant · LevEhad, retainer
Worked through infrastructure decisions in calls with developers and designers for an Israeli disability-inclusion NGO; contributed to the Google Cloud to AWS migration and a parallel hardware track.

2020 – 2022 — Research lead · Spring Research, quantitative research
Owned applied research on financial-market data end to end: monitoring and analysis, new signal representations and transformations, feature engineering, model selection, training; carried findings from exploratory analysis into the team's shared modelling.

2009 – 2019 — Data science, NLP and software roles · Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, Finfon
Financial-news NLP and signal extraction, medical-text classification and information extraction, computer-vision pipelines, corpus-linguistics infrastructure. Tel Aviv, Hong Kong, Prague.

STACK, EDUCATION, LANGUAGES

Languages. Rust, Elixir/OTP and Phoenix, Python, Swift, Clojure, TypeScript; earlier Haskell and Agda.

Machine learning. NLP, text classification and extraction, feature engineering, model selection and training; topological data analysis.

Models and routing. Claude, Gemini, OpenRouter, Ollama, Cloudflare Workers AI; tool calling, fallback, time and token budgets, latency tracing.

Education. MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Operations. Nix, Caddy, systemd, CI, authenticated HTTP APIs, credential-service design.

Spoken. English, Hebrew, Russian; working German and French.

Facts from `cv/knowledge-base.json`, 2026-11-06. Private repositories: walkthrough on a call.